

Smart Walking Stick Using Arduino UNO

Category: Assistive Technology | Status: Completed Prototype | 7 min read

An Arduino smart walking stick prototype that uses an HC-SR04 ultrasonic sensor, buzzer, and LED to detect nearby obstacles.

Materials

- Arduino UNO
- HC-SR04 ultrasonic sensor
- DC buzzer
- LED
- 9V battery with connector
- Jumper wires
- PVC pipe or walking stick
- Tape, glue, or mounting material

What I learned

- Assistive devices need user-centered thinking from the beginning.
- A working sensor is not enough if the alert is confusing or unreliable.
- Testing conditions matter, especially for safety-related prototypes.

Safety or accuracy note

This prototype is educational and should not replace certified mobility aids or professional support.